Curriculum Vitæ *updated: December 7, 2025*

Philipp M. Schicho

1. Personal information

name	Dr. Philipp Maximilian Schicho born in Graz, Austria, 01 October 1991
nationality	Austrian
address	26, Quai Ernest-Ansermet CH-1211 Genève 4, Switzerland
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2. Current position

09/2024 – present Research group leader (SNSF Ambizione fellow)
Thermal field theory, Particles, and Cosmic phases research group
Département de Physique Théorique, Université de Genève
Principle investigator of SNSF project no. 215997: *Gravitational waves from cosmological phase transitions: Precision cosmology in the light of LISA* (PRECLISA)

3. Employment history

09/2022 – 08/2024 Postdoctoral researcher
Institute for Theoretical Physics, Goethe University Frankfurt
Reference: L. Sagunski, J. Schaffner-Bielich

05/2020 – 08/2022 Postdoctoral researcher
Helsinki Institute of Physics, University of Helsinki
Reference: A. Vuorinen, K. Rummukainen

02/2017 – 04/2020 Doctor of Philosophy, PhD Physics (magna cum laude), 23/04/2020
AEC, Institute for Theoretical Physics, University of Bern
Advisor: M. Laine
Thesis: *Multi-loop investigations of strong interactions at high temperatures*, (cf. research output [4]).

10/2016 – 01/2017 Technical student
CERN, Accelerator and Beam Transfer, Beam Transfer Physics
Theoretical optimisation of slow extraction (cf. research output [6]).
Advisors: M.A. Fraser, M. Meddahi

06/2015 – 08/2015 Summer student
CERN, ABT, BTP
Thesis: *Optimising simulation times of SPS slow extraction using* `MAD-X`, (cf. research output [5]).

07/2014 – 08/2014 Summer student (GPA 1.0/1.0)
HEPHY, Institute of High Energy Physics, Vienna
Advisor: R. Schöfbeck
Thesis: *Increasing the sensitivity of a search for supersymmetry in the single lepton channel with the Stransverse Mass* M_{T2} (CMS), (cf. research output [1]).

4. Education

07/2017 Ècole de physique des Houches
Effective Field Theory (EFT) in particle physics and cosmology

03/2017 Computer algebra and particle physics (CAPP) school, DESY, Hamburg

09/2014 – 11/2016 Master of Science, MSc Physics (GPA 5.5/6.0), 01/11/2016
ETH Zürich, Switzerland
Major: Theoretical high energy physics, lattice QCD, applied mathematics
Advisor: P. de Forcrand
Thesis: *Inhomogeneous condensation in quark-based QCD effective models via wavelet pseudoparticles*, (cf. research output [3]).

07/2014 LAPP Annecy-le-Vieux, France
Summer School in Particle and Astroparticle physics

08/2013 Theoretical Physics Summer school, University of Utrecht, Netherlands

09/2011 – 08/2014 Bachelor of Science, BSc Physics (with distinction, GPA 1.1/1.0), 12/08/2014
Graz University of Technology, Austria
Advisors: H. G. Evertz, C. B. Lang
Thesis: *π- and ρ-Meson mass spectroscopy from Lattice QCD*, (cf. research output [2]).

09/2002 – 05/2010 Österreichische Reifeprüfung, Matura (with distinction, GPA 1.0/1.0)
AHS BG/BRG Leibnitz, Austria
Major: Physics and geometry
Advisor: H. Scherz

Thesis: *Sonoluminescence – A bubble's enlightenment.* A theoretical and experimental approach to the effect of Sonoluminescence.

5. Organization experience

08/2026 Conference organizer, *Strong and Electro-Weak Matter 2026*, University of Helsinki

05/2026 Workshop organizer, *Bubble Nucleation with Loop and Gravity Effects: Refining Gravitational-Waves Predictions*, IFPU (Trieste)

2025 – present Seminar organizer, Cosmology and particle physics seminars, University of Geneva

08/2025 Workshop organizer, *Advancing gravitational wave predictions from cosmological first-order phase transitions*, CERN

08/2024 Conference organizer, *Strong and Electro-Weak Matter 2024*, Goethe University Frankfurt

06/2024 Convener, *2nd Edition of the Workshop 'Quarkonia meet Dark Matter' (QMDM)*, Institute for Advanced Study, Technical University of Munich

01/2024 Conference organizer, *Mainz-Frankfurt-Münster CosmoCoffee*, Mainz

6. Outreach

09/2017 Public research display, University of Bern
Nacht der Forschung (NdF)

7. Professional services

2022 – present Referee
American Physical Society's journals: Phys. Rev. **D**, Phys. Rev. **L**
Springer's journals: JHEP, Eur. Phys. J. **C**

8. Scientific research skills

Theoretical (Dimensionally reduced) effective field theories, thermal field theory, quantum field theory, Lattice QCD, simulations in physics, computer algebra techniques, general relativity, cosmology, string theory, conformal field theory, group theory

Computational `C/C++`, `Python`, `Matlab`, `FORM`, `ROOT`, `FORTRAN 77`, Unix, Linux, Mathematica, LaTeX, computer hardware, `HTML`, Office, CAD-Software

9. Funding, awards, fellowships

08/2025 Awarded **CHF 20k** to co-organize the CERN TH institute *Advancing gravitational wave predictions from cosmological first-order phase transitions*

06/2025 Awarded best poster prize at *CosmoFondue*

2024 Awarded **CHF 810k** as PI of SNSF Ambizione Grant 215997, Swiss National Science Foundation (SNSF)

2011 – 2014 Scholarship of excellence Graz University of Technology
EUR 2.4k scholarship

10. Languages

German Mother-tongue

English Proficient C2, TOEFL 106/120 (2014), Cambridge ESOL B2 First FCE (2010)

Spanish Intermediate B1

Danish Elementary A2

French Beginner A2

Latin Very good (literal translation)

11. Major scientific achievements

11.1. Precision thermodynamics for cosmological phase transitions.

Invigorated by the first gravitational wave (GW) detections from binary mergers, I dedicated a large part of my research to improving the precision of thermodynamic predictions for cosmological phase transitions. To this end, I assessed theoretical uncertainties affecting such transitions in [12]. The motivation for this leap in precision was that leading-order (LO) thermodynamic calculations are insufficient, rendering GW predictions ambiguous. Even seemingly minor uncertainties in the early stages of the beyond the Standard Model (BSM) $\rightarrow$ GW pipeline can drastically alter predictions, potentially compromising successful stochastic gravitational wave background (SGWB) observations at LISA before they begin.

The work in [27] established the current state of the art in precision for GW predictions by completing perturbative computations for cosmological phase transitions. This level of accuracy was achieved by applying techniques of dimensional reduction to derive the three-dimensional SM EFT and by performing a three-loop computation of the effective potential.

11.2. Development of automated 3d EFT framework for thermal field theories.

To improve conceptual understanding of such EFT computations, we presented a didactic review [11] of thermal resummation techniques, combining perturbative and non-perturbative methods. Focusing on the dynamical real singlet extension of the Standard Model, I computed novel next-to-leading order (NLO) contributions to the parameters of the dimensionally reduced EFT. These results are intended to support future non-perturbative lattice studies.

We demonstrated that two-loop corrections in both matching and effective potential are substantial [26, 13] and essential for robust determinations of thermodynamic phase transition parameters. We also devised a minimal, gauge-invariant setup [17] that incorporates thermal resummation and supersedes previously used schemes.

In this context, I automated the evaluation and reduction of analytically challenging sum-integrals at non-zero temperature. This led to the development of the `Mathematica` package `DRalgo` [21], which has since been successfully applied not only in QCD but also in deriving the **most accurate predictions of gravitational waves** from cosmological first-order phase transitions in generic BSM theories.

11.3. Gauge-invariant framework for nucleation at finite temperature.

Theoretical uncertainties in determining the nucleation rate of bubbles of the stable vacuum at finite temperature remain a major challenge. In particular, radiatively induced transitions have long suffered from gauge-dependent ambiguities in the nucleation rate. I resolved this long-standing problem by employing effective field theory at the nucleation scale for the Abelian Higgs model. This allowed me to demonstrate, for the first time [18, 15], gauge invariance of the LO perturbative contributions in radiatively induced transitions.

11.4. Model-generic determination of the bubble wall velocity during first-order phase transitions.

A major milestone in improving the precision of GW predictions was the development of the public software package `WallGo` [29]. I contributed significantly to the software development. The code solves a coupled system of hydrodynamic equations, classical scalar equations of motion, and Boltzmann equations—including out-of-equilibrium effects—to determine the bubble wall velocity v_w for generic models. `WallGo` **is the first public tool capable of computing v_w with full inclusion of out-of-equilibrium contributions**.

12. List of publications

To date, I have authored 29 publications of which 23 are peer-reviewed in international journals and co-authored one review article. Altogether these works have accrued over 1200 citations (inspirehep.net, as of December 7, 2025), with a corresponding h-index of 18. Unless noted otherwise, the research list below is ordered alphabetically. For each item, I contributed at first- or second-author level: I carried out the computations, co-led the writing, and developed the main ideas. The complete publication list can also be found at Inspire HEP `inspirehep.net/authors/1639147`.

Journal articles

[33] J. van de Vis, **P. Schicho**, L. Niemi, B. Laurent, J. Hirvonen, and O. Gould, *WallGo investigates: Theoretical uncertainties in the bubble wall velocity*, (2025), [2510.27691].

[32] A. Bhatnagar, D. Croon, **P. Schicho**, *Interpreting the 95 GeV resonance in the Two Higgs Doublet Model: Implications for the Electroweak Phase Transition*, (2025), [2506.20716].

[31] M. Kierkla, N. Ramberg, **P. Schicho**, and D. Schmitt, *Theoretical uncertainties for primordial black holes from cosmological phase transitions*, (2025), [2506.15496].

[30] F. Bernardo, P. Klose, **P. Schicho**, and T. V. I. Tenkanen, *Higher-dimensional operators at finite temperature affect gravitational-wave predictions*, JHEP **08**, 109 (2025), [2503.18904].

[29] A. Ekstedt, O. Gould, J. Hirvonen, B. Laurent, L. Niemi, **P. Schicho**, and J. van de Vis, *How fast does the WallGo? A package for computing wall velocities in first-order phase transitions*, JHEP **04**, 101 (2025), [2411.04970].

[28] M. Kierkla, **P. Schicho**, B. Swiezewska, T. V. I. Tenkanen, and J. van de Vis, *Finite-temperature bubble nucleation with shifting scale hierarchies*, JHEP **07**, 153 (2025), [2503.13597].

[27] A. Ekstedt, **P. Schicho**, and T. V. I. Tenkanen, *Cosmological phase transitions at three loops: The final verdict on perturbation theory*, Phys. Rev. D **110**, 096006 (2024), [2405.18349].

[26] M. Lewicki, M. Merchand, L. Sagunski, **P. Schicho**, and D. Schmitt, *Impact of theoretical uncertainties on model parameter reconstruction from GW signals sourced by cosmological phase transitions*, Phys. Rev. D **110**, 023538 (2024), [2403.03769].

[25] J. Ghiglieri, **P. Schicho**, N. Schlusser, and E. Weitz, *The force-force correlator at the hard thermal scale of hot QCD*, JHEP **03**, 111 (2024), [2312.11731].

[24] C. Lwowski, K. Kaiser, J. Bucur, **P. Schicho**, and T. Kohnen, *Accuracy of using the axial length of the fellow eye for IOL calculation in retinal detachment eyes undergoing silicone oil removal*, Br. J. Ophthalmology (2023) `10.1136/bjo-2023-323581`.

[23] J. Österman, **P. Schicho**, and A. Vuorinen, *Integrating by parts at finite density*, JHEP **08**, 212 (2023), [2304.05427].

[22] L. Sagunski, **P. Schicho**, and D. Schmitt, *Supercool exit: Gravitational waves from QCD-triggered conformal symmetry breaking*, Phys. Rev. D **107**, 123512 (2023), [2303.02450].

[21] A. Ekstedt, **P. Schicho**, and T. V. I. Tenkanen, `DRalgo`*: A package for effective field theory approach for thermal phase transitions*, Comput. Phys. Commun. **288**, 108725 (2023), [2205.08815].

[20] T. Gorda, A. Kurkela, J. Österman, R. Paatelainen, S. Säppi, **P. Schicho**, K. Seppänen, and A. Vuorinen, *Degenerate fermionic matter at N3LO: Quantum electrodynamics*, Phys. Rev. D **107**, L031501 (2023), [2204.11893].

[19] T. Gorda, A. Kurkela, J. Österman, R. Paatelainen, S. Säppi, **P. Schicho**, K. Seppänen, and A. Vuorinen, *Soft photon propagation in a hot and dense medium to next-to-leading order*, Phys. Rev. D **107**, 036012 (2023), [2204.11279].

[18] J. Löfgren, M. J. Ramsey-Musolf, **P. Schicho**, and T. V. I. Tenkanen, *Nucleation at Finite Temperature: A Gauge-Invariant Perturbative Framework*, Phys. Rev. Lett. **130**, 251801 (2023), [2112.05472].

[17] **P. Schicho**, T. V. I. Tenkanen, and G. White, *Combining thermal resummation and gauge invariance for electroweak phase transition*, JHEP **11**, 047 (2022), [2203.04284].

[16] S. Biondini, **P. Schicho**, and T. V. I. Tenkanen, *Strong electroweak phase transition in t-channel simplified dark matter models*, JCAP **10**, 044 (2022), [2207.12207].

[15] J. Hirvonen, J. Löfgren, M. J. Ramsey-Musolf, **P. Schicho**, and T. V. I. Tenkanen, *Computing the gauge-invariant bubble nucleation rate in finite temperature effective field theory*, JHEP **07**, 135 (2022), [2112.08912].

[14] J. Ghiglieri, G. D. Moore, **P. Schicho**, and N. Schlusser, *The force-force-correlator in hot QCD perturbatively and from the lattice*, JHEP **02**, 58 (2022), [2112.01407].

[13] L. Niemi, **P. Schicho**, and T. V. I. Tenkanen, *Singlet-assisted electroweak phase transition at two loops*, Phys. Rev. D **103**, 115035 (2021), [2103.07467].

[12] D. Croon, O. Gould, **P. Schicho**, T. V. I. Tenkanen, and G. White, *Theoretical uncertainties for cosmological first-order phase transitions*, JHEP **04**, 055 (2021), [2009.10080].

[11] **P. M. Schicho**, T. V. I. Tenkanen, and J. Österman, *Robust approach to thermal resummation: Standard Model meets a singlet*, JHEP **06**, 130 (2021), [2102.11145].

[10] M. Laine, **P. Schicho**, and Y. Schröder, *A QCD Debye mass in a broad temperature range*, Phys. Rev. D **101**, 023532 (2020), [1911.09123].

[9] M. Laine, **P. Schicho**, and Y. Schröder, *Soft thermal contributions to 3-loop gauge coupling*, JHEP **2018**, 37 (2018), [1803.08689].

Conference proceedings

[8] J. Ghiglieri, G. D. Moore, **P. Schicho**, N. Schlusser, and E. Weitz. *Hard parton dispersion in the quark-gluon plasma, non-perturbatively.* In *11th International Conference on Hard and Electromagnetic Probes of High-Energy Nuclear Collisions* (July 2023). [2307.09297].

[7] G. Aarts, J. Aichelin, C. Allton, A. Athenodorou, D. Bachtis, C. Bonanno, N. Brambilla, E. Bratkovskaya, M. Bruno, M. Caselle, C. Conti, R. Contino, L. Cosmai, F. Cuteri, L. Del Debbio, M. D'Elia, P. Dimopoulos, F. Di Renzo, T. Galatyuk, J. N. Guenther, R. Houtz, F. Karsch, A. Y. Kotov, M. P. Lombardo, B. Lucini, L. Maio, M. Panero, J. M. Pawlowski, A. Pelissetto, O. Philipsen, A. Rago, C. Ratti, S. M. Ryan, F. Sannino, C. Sasaki, **P. Schicho**, C. Schmidt, S. Sharma, O. Soloveva, M. Sorba, and U.-J. Wiese. *Phase Transitions in Particle Physics – Results and Perspectives from Lattice Quantum Chromo-Dynamics.* In (Jan. 2023). [2301.04382].

[6] M. Fraser, D. Björkman, K. Cornelis, B. Goddard, V. Kain, **P. Schicho**, C. Theis, and H. Vincke. *Modelling the Radioactivity Induced by Slow-Extraction Losses in the CERN SPS.* In *Proc. of International Particle Accelerator Conference (IPAC'17)* (May 2017), 1897–1900.

[5] M. A. Fraser, R. G. Alia, B. Balhan, H. Bartosik, C. Bertone, D. Björkman, J. Borburgh, N. Conan, K. Cornelis, L. Gatignon, B. Goddard, Y. Kadi, V. Kain, A. Mereghetti, F. Roncarolo, **P. M. Schicho**, J. Spanggaard, O. Stein, L. Stoel, F. M. Velotti, and H. Vincke. *SPS Slow Extraction Losses and Activation: Challenges and Possibilities for Improvement.* In *Proc. of International Particle Accelerator Conference (IPAC'17)* (Copenhagen. 2017), 611–614.

Theses

[4] **P. M. Schicho**, *Multi-loop investigations of strong interactions at high temperatures*, PhD thesis (U. Bern, 2020).

[3] **P. M. Schicho**, *Inhomogeneous condensation in quark-based QCD effective models via wavelet pseudoparticles*, MA thesis (ETH Zürich, 2016).

[2] **P. Schicho**, *π- and ρ-Meson mass spectroscopy from Lattice QCD*, BA thesis (TU Graz, 2014).

[1] **P. Schicho**, *Increasing the sensitivity of a search for supersymmetry in the single lepton channel with the Stransverse Mass*, Project thesis (HEPHY Vienna, 2014).

13. List of presentations

Active participation as a speaker in the most important international conferences and workshops in the fields of particle cosmology, thermal QCD, and phase transitions. Since 2017, I have delivered a total of ∼10 invited conference talks, ∼15 contributed conference talks, and ∼30 seminars.

13.1. Invited talks at conferences and workshops

27/09/2025 *Perturbative cosmological phase transitions in a broad temperature range*, invited **plenary** talk at *The 2025 Beijing Particle Physics and Cosmology Symposium (BPCS 2025)*, Beijing, China

04/08/2025 *Perturbative cosmological phase transitions in a broad temperature range*, invited **plenary** talk at *Numerical simulations of early Universe sources of gravitational waves* at NORDITA, Stockholm, Sweden

05/03/2025 *Limits and limitations of perturbative approaches to cosmological phase transitions*, invited talk at GW day at CERN, Geneva, Switzerland

08/10/2024 *Reliable theoretical predictions for GWs from first-order phase transitions*, invited **plenary** talk at *Fun with the phases of the Universe* at IFPU, Trieste, Italy

20/09/2024 *Reliable theoretical predictions for GWs from first-order phase transitions*, invited **plenary** talk at *Fundamental physics and gravitational wave detectors* at Pollica physics center, Italy

19/03/2024 *Strong cosmological phase transitions through the EFT lens*, invited **plenary** talk at *Quarkonia meet Dark Matter* at Institute for Advanced Study, Technical University of Munich, Germany

05/12/2022 *EFT framework for (precision) cosmological phase transition thermodynamics*, invited **plenary** talk at *What the heck happens when the Universe boils?* at Kavli IPMU, Tokyo, Japan

13.2. Contributed talks

16/09/2025 *Connecting micro- and intermediate scales in perturbative cosmological phase transitions*, contributed talk at Swiss LISA theory group meeting, UZH Zürich, Switzerland

06/06/2025 *Higher-dimensional operators at finite-temperature affect gravitational-wave predictions*, contributed talk at Swiss Cosmology Days 2025, ETH Zürich, Switzerland

05/06/2025 *Higher-dimensional operators at finite-temperature affect gravitational-wave predictions*, contributed talk at HEFT 2025, Geneva, Switzerland

21/02/2025 *Completing the perturbative program for cosmological phase transitions*, contributed talk at KEK theory meeting on particle physics phenomenology (KEK-PH2025winter), Tsukuba, Japan

04/09/2024 *Finite-temperature effective theories for first-order cosmological phase transitions*, contributed talk at High-Energy Theoretical Particle Physics Mini-Conference, Genève, Switzerland

14/09/2023 *Phase transition thermodynamic parameters at high precision*, contributed talk at 26th International Conference on Particle Physics and Cosmology (COSMO'23), Madrid, Spain

11/07/2023 *Degenerate fermionic matter at N^3LO*, contributed talk at ELEMENTS Annual Conference, Bad Nauheim, Germany

03/07/2023 *Can EFT reveal if there was an electroweak phase transition?*, contributed talk at Cosmology from Home

22/06/2023 `PRECLISA`. *Gravitational waves from cosmological phase transitions: Precision cosmology in the light of LISA*, interview talk for the Ambizione grant of the Swiss National Science Foundation, Bern, Switzerland

17/05/2023 *Phase transition thermodynamic parameters at high precision*, contributed talk at How fast does the bubble grow?, DESY, Hamburg, Germany

05/05/2023 *Strong electroweak phase transition and simplified dark matter models*, contributed talk at Progress on Old and New Themes in cosmology (PONT) 2023, Avignon, France

20/06/2022 *Jet dispersion in hot QCD from the lattice*, contributed talk at SEWM 2022, Paris, France

06/04/2022 *(Non-)perturbative jet dispersion hot QCD*, contributed talk at Quark Matter 2022, Kraków, Poland

30/03/2022 *(Non-)perturbative jet dispersion hot QCD*, contributed talk at Mini workshop: Phase transitions in particle physics, Galileo Galilei Institute, Firenze, Italy

29/03/2021 *Soft thermal contributions to 3-loop gauge coupling*, contributed parallel talk at FunQCD (online), Barcelona, Spain

28/06/2018 *Fun with thermal dimension-six operators*, contributed parallel talk at SEWM 2018, Barcelona, Spain

13.3. Seminar talks

20/11/2025 *Uncertainty budgets for bubble wall velocities in cosmological phase transitions*, invited seminar talk at Marietta Blau Institute for Particle Physics ÖAW, Vienna, Austria

23/09/2025 *Uncertainty budgets for bubble wall velocities in cosmological phase transitions*, invited seminar talk at ITP CAS, Beijing, China

27/03/2025 *Limits and limitations of perturbative approaches to cosmological phase transitions*, invited seminar talk at Warsaw University, Poland

26/02/2025 *Perturbative cosmological phase transitions and their bubble wall velocities*, invited seminar talk at Kavli IPMU, Tokyo, Japan

10/06/2024 *Completing the perturbative program for cosmological phase transitions*, invited seminar talk at Bielefeld University, Germany

23/11/2023 *Impact of computational diligence on GW signals from phase transitions*, invited seminar talk at Warsaw University, Poland

12/10/2023 *Impact of computational diligence on GW signals from phase transitions*, invited seminar talk at University of Basel, Basel, Switzerland

19/09/2023 *Impact of computational diligence on GW signals from phase transitions*, invited seminar talk at Helsinki Institute of Physics, Helsinki, Finland

15/06/2023 *Electroweak phase diagram at finite lepton number density*, contributed talk at Thermal Field Theory meets Phenomenology, Uppsala, Sweden

30/05/2023 *High-temperature effective field theories: Precision phase transition thermodynamics*, seminar talk at HEP seminar, Université de Genève, Switzerland

10/05/2023 *Integrating by parts at finite density*, informal talk at the DMGW meeting, Goethe University, Frankfurt, Germany

25/01/2023 *Degenerate fermionic matter at N^3LO*, invited seminar talk at Gravitation and Cosmology seminar, Utrecht University, Netherlands

24/01/2023 *What can EFT tell us about the electroweak phase transition?*, seminar talk at CRC-TR211 meeting and Colloquium, Bielefeld University, Germany

08/11/2022 *EFT framework for cosmological phase transition thermodynamics*, seminar talk at the AstroCoffee, Goethe University, Frankfurt, Germany

13/10/2022 *Degenerate fermionic matter at N^3LO*, invited seminar talk (online) at S@INT seminar, INT, Seattle, USA

15/09/2022 *(Gauge independent) Bubble nucleation rate at finite temperature*, invited seminar talk at Jožef Stefan Institute, Ljubljana, Slovenia

24/08/2022 *Can EFT tell us if there was an electroweak phase transition?*, invited seminar talk at University of Graz, Graz, Austria

11/07/2022 *Soft light-cone observables from electrostatic QCD*, invited seminar talk (online) at the QCD theory seminar

07/07/2022 *Degenerate fermionic matter at N^3LO*, invited seminar talk at the Nuclear Physics Colloquium, Goethe University, Frankfurt, Germany

16/06/2022 *Can EFT tell us if there was an electroweak phase transition?*, invited seminar talk at SUBATECH, Nantes, France

31/05/2022 *Electroweak phase transition: Combining thermal resummation and gauge invariance*, invited seminar talk at NICPB, Tallinn University, Estonia

24/05/2022 *Combining thermal resummation and gauge invariance for electroweak phase transition*, invited seminar talk (online) at School of Physics and Astronomy, Monash University, Australia

03/03/2022 *Effective theory approach to cosmological phase transitions*, invited seminar talk at Instituto de Astrofísica de Canarias, La Laguna, Spain

28/10/2021 *Gauge independent bubble nucleation rate at finite temperature*, invited seminar talk at University of Basel, Basel, Switzerland

19/10/2021 *Cosmological phase transition: Robust thermal resummation*, invited seminar talk at University of Bern, Bern, Switzerland

13/05/2021 *Cosmological phase transition: Robust thermal resummation*, invited seminar talk (online) at KIAS, Seoul, South Korea

25/11/2020 *How to be precise at the electroweak scale at finite-temperature*, invited seminar talk (online) at Kavli IPMU, Tokyo, Japan

13/08/2019 *3-Loop Gauge Coupling in Hot Yang-Mills*, invited seminar talk at Helsinki Institute of Physics, Helsinki, Finland

28/08/2018 *Fun with thermal dimension-six operators*, invited seminar talk at Universidad del Bío-Bío, Chillán, Chile

14. Teaching activities

14.1. Lecturer

02/2025 *Bubble wall velocities in cosmological phase transitions* 1wk×5hr
Yonsei-Konkuk-Sogang mini-workshop on first-order phase transitions

10/2023 – 02/2024 Institute for Theoretical Physics, Goethe University Frankfurt
Theoretische Physik 1: Mathematische Ergänzungen 15wk×2hr

03/2022 *Phase transitions in the early universe* (exercises) 1wk×5hr
Galileo Galilei Institute for Theoretical Physics
Theoretical Aspects of Astroparticle Physics, Cosmology and Gravitation

14.2. Teaching assistant

2022 Institute for Theoretical Physics, Goethe University Frankfurt
Astrophysics II

2017 – 2020 AEC, Institute for Theoretical Physics, University of Bern
Standard Model, *Quantum mechanics I*, *Quantum mechanics II*, *Exercises in theoretical physics*, *Specialist course: Introduction to BSM physics*, *Statistical thermodynamics II*

2015/16 ETH Zürich, D-MATH/D-PHYS
Numerical analysis I, *Numerical methods*, *Physics II*

2013/14 Graz University of Technology, ITP/IEP
Theoretical mechanics, physics laboratory I/II

14.3. Supervision

14.3.1. PhD students as primary supervisor

2024 – present F. Bernardo (U. Geneva)

14.3.2. PhD theses mentor

2022 – 2024 R. Diedrichs, (U. Frankfurt) supervised by L. Sagunski

2022 – 2024 D. Schmitt, (U. Frankfurt) supervised by L. Sagunski

2020 – 2024 J. Österman, (U. Helsinki) supervised by A. Vuorinen

14.3.3. Master students as primary or joint primary supervisor

2025/26 R. G. Reinle, (ETH Zürich)

2025/26 S. Kämpf, (U. Geneva)

2021/22 S. Vihko, (U. Helsinki) co-supervised with A. Vuorinen
Thesis: *EFT methods and calculational techniques in imaginary time formalism of thermal QCD.*

14.3.4. Bachelor students as primary or joint primary supervisor

2022/23 R. Baumann, (U. Frankfurt) co-supervised with L. Sagunski, D. Schmitt
Thesis: *Gravitational waves in the Abelian Higgs model*

14.3.5. Project theses

2024/25 B. Adadi, (ETH Zürich)